

Phuong Anh Lam

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Education

Case Western Reserve University, Cleveland, OH

Expected May 2031

Ph.D in Chemistry

Research Advisor: Dr. Anna Samia

Augustana College, Rock Island, IL

May 2022

Bachelor of Arts in Chemistry and Communication Studies

GPA: 3.51

Publication

1. Nguyen, T. H.; Tran, N. A.; Nguyen, T. B. N.; Bui, T. M. D.; **Lam, P. A.**; Choi, H.-S.; Do, D. L.; Nguyen, T. H.; Hoang, V. P.; Nguyen, H. L.; Pham, T. D.; Dao, V.-D. Carbonized Rice Husks as Absorber Materials for Solar-Driven Steam Generation Devices. *Diamond and Related Materials* **2026**, 161, 113083.

2. Ngoc, N. T. B.; Hoang, N. M.; Trang, V. T.; Tu, L. T.; Mai, T. Q.; Tran Vu, H.-A.; Ho, N. X.; Vu, H.-H. T.; Doan, V. A.; **Lam, P. A.**; Dao, V.-D. Magnetic Field Role in Solar-Driven Water Evaporation of Fe₃O₄/Polyvinyl Alcohol. *RSC Advances* **2025**, 15 (58), 49841–49851.

Research Experience

Graduate Researcher

Sep 2025 – Present

Samia Lab, Case Western Reserve University, Cleveland, OH

- Synthesized green TiO₂ nanoparticles via ball milling methods for photocatalysis and antibacterial applications.
- Investigated reactive oxygen species (ROS) generation under LED and visible light irradiation using fluorescence spectrophotometry; evaluated photocatalytic performance by correlating irradiation conditions with ROS production.
- Developed carbon dots via hydrothermal synthesis methods with different purification techniques for photothermal and antibacterial applications.

Research Associate

May 2023 – Oct 2024

KEA Technologies, Inc., Littleton, MA

- Delivered two major projects: Driver Alcohol Detection System for Safety (DADSS), a research program funded by the U.S. Department of Transportation that develops an in-vehicle non-dispersive infrared (NDIR) alcohol detection system to help eliminate drunk driving, and Breath Alcohol Ignition Interlock Device (BAIID), an internal project focused on developing a commercial breathalyzer system.
- Designed and performed accelerated testing procedures to investigate environmental effects (temperature, humidity, VOCs) on breath alcohol sensors using FTIR.
- Applied electrochemical impedance spectroscopy (EIS) to study polymer electrolyte membrane (PEM) fuel cell degradation under environmental stressors (temperature, humidity).
- Developed UPLC-PDA methods to quantify VOC leachables from polymer materials.
- Co-led infrared imaging studies to analyze breath flow dynamics using MATLAB.
- Analyzed and visualized data using R and Excel; delivered weekly presentations and quarterly technical reports on methods and results.

- Mentored 2 interns; provided training and guidance on instrumentation, method development, and experiment troubleshooting.

Undergraduate Researcher

June 2021 – Nov 2021

Biggin Lab, Augustana College, Rock Island, IL

- Developed HPLC-UV/Vis methods to quantify cannabinoids (CBD, CBG) and impurities (hexane, benzene) in commercial products.
- Optimized sample preparation and chromatographic conditions (mobile phase, column type, flow rate, heating temperature).
- Analyzed data and presented findings to faculty and peers.

Teaching Experience

Graduate Teaching Assistant

Aug 2025 – Present

Case Western Reserve University, Cleveland, OH

- Graded exams and weekly assignments for undergraduate chemistry courses (Principles of Chemistry I, II; Concepts for a Molecular View of Biology I, II; Introductory Organic Chemistry I, II; Organic Chemistry I).
- Assisted instructors with exam administration and assessment.
- Evaluated student performance and provided constructive feedback to support student learning.

Undergraduate Teaching Assistant

Sept 2021 – May 2022

Augustana College, Rock Island, IL

- Supervised 15 students doing experiments, troubleshooting, and writing laboratory reports (General Chemistry I and Analytical Chemistry)
- Supported laboratory setup, inventory management, and equipment maintenance.
- Ensured students follow laboratory safety protocols; responded to laboratory incidents.

Technical Skills

Analytical Techniques: FTIR, UPLC-PDA, HPLC-UV/Vis, ICP-OES, AAS, EIS, scanning electron microscopy, fluorescence spectrophotometry.

Computational & Data Analysis: Python, MATLAB, R; data processing, visualization, and basic automation of experimental workflows.

Laboratory Skills: Experimental design, method development, instrument calibration and troubleshooting, chemical safety and hazardous material handling.

Relevant Experience

Laboratory Manager

Jan 2025 – June 2025

Black Hawk College, Moline, IL

- Assisted with experimental design, setup, and troubleshooting across chemistry, biology, and engineering courses.
- Maintained and troubleshot over 20 laboratory equipment; ensured compliance with institutional and safety regulations.
- Managed laboratory inventory and budgeting for chemicals and equipment.
- Trained and supervised new laboratory personnel in instrument operation and chemical safety protocols.

Analytical Chemist

June 2022 – May 2023

Q.C. Metallurgical Laboratory, Inc., Davenport, IA

- Performed trace metal analysis using ICP-OES and flame atomic absorption spectroscopy (AAS) on environmental and industrial samples.
- Conducted material characterization and failure analysis using FTIR and scanning electron microscopy (SEM).
- Carried out water quality testing (BOD, COD, TSS, TDS, *E.coli* testing).
- Contributed to maintaining ISO 17025:2017 accreditation through standardized procedures and documentation.
- Trained and supervised an intern on daily calibration, product testing, and data analysis.

Cosmetic Chemist

June 2021 – May 2022

Argrow's House, Davenport, IA

- Developed sustainable cosmetic products; expanded color and scent collections to meet diverse consumer preferences.
- Conducted quality control testing including pH and viscosity measurements.
- Collaborated with the marketing team and the local news channel, KWQC, to promote sustainable products and expand the online customer base.

Assistant Teacher

Feb 2022 – May 2022

Pleasant View Elementary School, Bettendorf, IA

- Provided language translation sessions and multilingual teaching materials to help Vietnamese students build confidence in learning science in English.
- Developed small-group activities to encourage student participation and social skills.
- Attended monthly parent-teacher conferences to provide updates on the students' progress and challenges.

Science Communicator

Mar 2019 – Aug 2021

Junie B, Facebook, Instagram

- Founded a cross-channel media platform that reached over 5000 members to raise awareness on skin health and cancer prevention among Vietnamese youth.
- Produced educational content to explain the chemistry behind personal care products.
- Collaborated with industry professionals to discuss cosmetic ingredient safety and effectiveness.

Awards

Summer Student Research Fellowship, Augustana College, Department of Chemistry, 2021.**Augie Choice Grant**, Augustana College, 2021.**Summer Servant Leader Internship Grant Program**, Augustana College, 2021.